Rodrigo Bazan

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Education

UNIVERSITY OF FLORIDA May 2026
B.S. in Computer Engineering | GPA: 3.56 *Gainesville, FL*
Relevant Coursework: Web Development, Software Engineering, Data Structures & Algorithms, Operating Systems
Leadership: IEEE Student Branch – Community Service Lead managing event logistics for 150+ ECE students

Experience

ASSOCIATE ENGINEER I Aug. 2026 – Present
ITC Holdings Corp. *Novi, MI*

- Deployed a **Python** desktop tool with a **Tkinter** GUI to search 6,000+ protection settings by substation and ID, automating a 16-step manual file-retrieval workflow into a single click for a 10-person engineering team.
- Integrated Excel read/write (**openpyxl**), file copying and relay software launches into a single interface, eliminating manual navigation across 3 separate applications for every settings update.
- Saved 6+ hours of manual work by extending the desktop tool's search logic into a Python script (**pathlib, shutil, re**) that deduplicated, validated, and migrated 3,000+ legacy files into a new directory structure with zero errors.
- Cut fault-simulation test time by 40% by writing macros that query a Firebird database via **SQL** to extract relay data and calculate simulation parameters automatically.

RELAY DESIGN INTERN May 2025 – Aug. 2025
ITC Holdings Corp. *Novi, MI*

- Built a **Python** data pipeline with **pandas** to parse and reconcile 1,000+ records across field and simulation datasets, flagging 30+ discrepancies so the team could update simulation models to match real-world equipment.
- Reduced design report turnaround time by 20% by engineering few-shot prompt templates that converted plain-English engineering notes into standardized report drafts.

SOFTWARE TEST ENGINEER INTERN May 2024 – Aug. 2024
Paradox Security Systems *Fort Lauderdale, FL*

- Improved inventory lookup efficiency by 25% by designing a **SQL** database with a **Python** interface to catalog firmware version, hardware type, production year, and storage location for 200+ products.
- Designed reusable LLM prompt templates to generate 75+ structured test inputs in CSV, expanding QA coverage across 4 product configurations and cutting test prep time by 40%.

Projects

PICKLEBALL TOURNAMENT APP | *Flask, Python, HTML/CSS* Jun. 2026

- Deployed a full-stack Flask web app on Render.com that ran a live, custom-formatted tournament for 40+ users.
- Implemented player registration, team drafting, live score entry, and Python-computed team rankings/standings.
- Designed a mobile-friendly experience enabling a shared tournament state accessible via participants' phones.

ATHLETE POSITION TRACKING SYSTEM | *ESP32, C++, Python, Flask* Apr. 2026

- Created a wearable position tracking system using ESP32s (C++) and trilateration algorithms for amateur athletes.
- Analyzed 1,000+ distance points between anchors to trilaterate precise player coordinates within 10 cm accuracy.
- Transformed raw data into heatmap visualizations on a web-based GUI running the Python algorithms via Flask.

NBA FAN APP | *React, ESPN API, Node.js, MySQL* Apr. 2025

- Built a full-stack sports application with a React frontend integrated with the ESPN API to display live NBA scores.
- Implemented user authentication, favorites tracking, and search functionality using Node.js and MySQL.
- Collaborated with a 5-person team via Git workflows to design and deploy a responsive, engagement-focused UI.

Skills

Programming Languages: Python, Java, C/C++, SQL, JavaScript, HTML/CSS
Tools & Frameworks: React, Tailwind, MySQL, Git, Flask, Node.js, REST APIs, Linux, Claude Code